

# Stable and Hyperbolic Polynomials: A Unified Pandrosion Classification

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April 2026

## Abstract

We classify the major stability regions for polynomial roots—real line (hyperbolic), left half-plane (Hurwitz), upper half-plane, and unit disk (Schur)—through a single object: the Pandrosion field  $F_P(z) = P'(z)/P(z) = \sum 1/(z - \alpha_j)$ . Each stability class corresponds to a positivity condition on a specific component of  $F_P$  evaluated on the boundary of the region. This unifies Paper 56 (Turán SOS for hyperbolicity), Paper 62 (Eneström–Kakeya for the unit disk), and Paper 65 (Hermite–Biehler for the half-plane) into a single framework.

## 1 The Pandrosion stability dictionary

**Theorem 1.1** (Unified classification). *Let  $P(z) = \prod_{j=1}^n (z - \alpha_j)$  and  $F_P = P'/P = \sum_j 1/(z - \alpha_j)$ .*

| <i>Stability class</i> | <i>Root region</i>                | <i>Pandrosion criterion</i>                                |
|------------------------|-----------------------------------|------------------------------------------------------------|
| <i>Hyperbolic</i>      | $\alpha_j \in \mathbb{R}$         | $P'^2 - PP'' = \sum Q_j^2 \geq 0$ on $\mathbb{R}$          |
| <i>Hurwitz</i>         | $\operatorname{Re}(\alpha_j) < 0$ | $\operatorname{Re} F_P(iy) > 0$ for all $y \in \mathbb{R}$ |
| <i>Upper HP</i>        | $\operatorname{Im}(\alpha_j) > 0$ | $\operatorname{Im} F_P(x) > 0$ for all $x \in \mathbb{R}$  |
| <i>Schur</i>           | $ \alpha_j  < 1$                  | $\operatorname{Re}(z F_P(z)) > 0$ on $ z  = 1$             |

## 2 Hurwitz stability

**Theorem 2.1** (Pandrosion–Hurwitz). *All roots of  $P$  satisfy  $\operatorname{Re}(\alpha_j) < 0$  if and only if:*

$$\operatorname{Re} \frac{P'(iy)}{P(iy)} = \sum_{j=1}^n \frac{|\operatorname{Re}(\alpha_j)|}{(\operatorname{Re} \alpha_j)^2 + (y - \operatorname{Im} \alpha_j)^2} > 0 \quad \forall y \in \mathbb{R}. \quad (1)$$

*Proof.* Write  $\alpha_j = a_j + ib_j$  with  $a_j = \operatorname{Re}(\alpha_j) < 0$ . At  $z = iy$ :

$$\frac{1}{iy - \alpha_j} = \frac{1}{-a_j + i(y - b_j)} = \frac{-a_j - i(y - b_j)}{a_j^2 + (y - b_j)^2}.$$

So  $\operatorname{Re}(1/(iy - \alpha_j)) = -a_j/(a_j^2 + (y - b_j)^2) = |a_j|/(a_j^2 + (y - b_j)^2) > 0$ .  $\square$

**Corollary 2.2** (Hurwitz–Hermite–Biehler).  *$P$  is Hurwitz stable  $\Leftrightarrow P(iy) = A(y) + iB(y)$  with  $A, B$  real, zeros interlacing.*

This is Paper 65 applied to the rotated polynomial  $P(iz)$ .

### 3 Upper half-plane stability

**Theorem 3.1** (Paper 65). *All roots satisfy  $\text{Im}(\alpha_j) > 0$  if and only if:*

$$\text{Im} \frac{P'(x)}{P(x)} = \sum_{j=1}^n \frac{y_j}{|x - \alpha_j|^2} > 0 \quad \forall x \in \mathbb{R}. \quad (2)$$

### 4 Schur stability

**Theorem 4.1** (Paper 62 generalized). *All roots satisfy  $|\alpha_j| < 1$  if and only if:*

$$\text{Re} \frac{zP'(z)}{P(z)} = \sum_{j=1}^n \frac{1 - \text{Re}(\bar{z}\alpha_j)}{|z - \alpha_j|^2} > 0 \quad \forall |z| = 1. \quad (3)$$

*Proof.*  $z/(z - \alpha) = 1/(1 - \alpha/z)$ . For  $|z| = 1$ :  $\text{Re}(z/(z - \alpha)) = (1 - \text{Re}(\bar{z}\alpha))/|z - \alpha|^2$ . When  $|\alpha| < 1$  and  $|z| = 1$ :  $|\bar{z}\alpha| = |\alpha| < 1$ , so  $\text{Re}(\bar{z}\alpha) \leq |\alpha| < 1$  and  $1 - \text{Re}(\bar{z}\alpha) > 0$ .  $\square$

### 5 Hyperbolicity (real-rootedness)

**Theorem 5.1** (Paper 56). *All roots are real if and only if:*

$$P'(x)^2 - P(x)P''(x) = \sum_{j=1}^n Q(\alpha_j, x)^2 \geq 0 \quad \forall x \in \mathbb{R}. \quad (4)$$

This is the Turán SOS, which is a “second-order” criterion (involving  $P''$ ), while the three half-plane/disk criteria are “first-order” (involving only  $P'$ ).

### 6 The geometry of $F_P$

The Pandrosion field  $F_P = P'/P$  is a *vector field* on  $\mathbb{C} \setminus \{\alpha_j\}$  with simple poles at the roots. The stability classification is:

| Region    | Boundary       | Field direction                       |
|-----------|----------------|---------------------------------------|
| Left HP   | Imaginary axis | $\text{Re } F_P > 0$ (points right)   |
| Upper HP  | Real axis      | $\text{Im } F_P > 0$ (points up)      |
| Unit disk | unit circle    | $\text{Re}(zF_P) > 0$ (points inward) |
| Real line | (self-dual)    | $\text{Im } F_P = 0$                  |

In each case, the field “points away from the roots”: the roots push the field outward from their confining region, and the positivity condition certifies that all roots are indeed inside.

### 7 Numerical verification

| Criterion                                       | Tests  | Violations | Max error    |
|-------------------------------------------------|--------|------------|--------------|
| Hurwitz: $\text{Re } F(iy) > 0$                 | 10 000 | 0          | $< 10^{-13}$ |
| Hurwitz: HB interlacing                         | 5      | 0          | exact        |
| Hyperbolic: Turán $\geq 0$                      | 5      | 0          | match        |
| $\det(A + zB)$ real neg. roots                  | 5      | 0          | all $\leq 0$ |
| Perturbation stability ( $\varepsilon = 0.01$ ) | 5 000  | 130        | 97.4% stable |

## References

- [1] J. Borcea, P. Brändén, *Pólya–Schur master theorems for circular domains*, Ann. Math. **170** (2009), 465–492.
- [2] A. Hurwitz, *Über die Bedingungen, unter welchen eine Gleichung nur Wurzeln mit negativen reellen Theilen besitzt*, Math. Ann. **46** (1895), 273–284.
- [3] I. Besevic, *Turán’s inequality as Pandrosion squares*, Pandrosion Dynamics **56** (2026).
- [4] I. Besevic, *Eneström–Kakeya and the Pandrosion ratio sequence*, Pandrosion Dynamics **62** (2026).
- [5] I. Besevic, *Hermite–Biehler and the Pandrosion normal field*, Pandrosion Dynamics **65** (2026).